

SPATIAL AND TEMPORAL ANALYSIS OF LAND COVER AND LAND SURFACE TEMPERATURE IN GCR+ FROM 2017-2023

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ABSTRACT:

The Philippines experiences not only more frequent and stronger typhoons but also abnormally high temperatures. Because of this, it is imperative that studies on the subject are taken to understand the necessary actions we are required to take. GIS is an indispensable for the analysis of land surface temperature (LST) and land cover change. It can enable researchers to analyze and map the spatial distribution of temperature and land cover. This study focuses on exploring the changes that have happened in the land cover and surface temperature of GCR+. This study also aims to look at the relationship between these two variables. The results of this study showed that tree cover and vegetation, in general, is still the most dominant land cover type across the region. However, tree cover and vegetation are also currently experiencing steady and gradual decline. Laguna, Rizal, and Bulacan have the highest vegetation cover and are also the provinces that experienced the lowest temperatures. On the other hand, NCR, Cavite, and Pampanga are the three provinces with the highest temperatures and likewise the top three in total urban area percentage. In terms of temperature trends, this study observed that there was a sharp decrease in temperature in 2020 and 2021 and an abrupt increase in temperature post-2021. It was also observed that built-up areas and bare soil generally exhibit higher temperatures or positive temperature change while trees exhibit the opposite. Finally, this study concludes that changes in land cover do not completely explain variations in temperature; however, general trends can be observed.

1. INTRODUCTION

1.1 Background of the Study

1.1.1 Climate Change and Global Warming

According to the United Nations (n.d.), climate change is the long-term shifts of the Earth's temperatures and weather patterns. These shifts in temperature happen naturally because of the changes in the sun's activity and, in some cases, large-scale volcanic activities. However, human activities have been the main driver and catalyst of climate change in modern times (United Nations, n.d.). Activities such as burning of coal and fossil fuels release large amounts of greenhouse gases in the atmosphere. These gases are very effective in trapping heat causing a rise in temperature. A change in global temperature as small as 1°C is already significant in increasing the frequency and magnitude of the most extreme weather-related events such as droughts, hurricanes, flooding, and wildfires (UNEP, n.d.).

Philippines is one of the countries of the world classified with severe vulnerability to climate change (National Geographic, 2011). Because of this, Philippines experiences not only more frequent and stronger typhoons but also higher than normal temperatures. In the month of April 2024, a series of class suspensions were made by schools citing extreme heat and possible health issues that may arise from this. For example, on April 12, 2024 alone, the Department of Education (DepEd) reported that a total of 7,080 schools suspended in-person classes and shifted to remote delivery modes that day (Rita, 2024). These significant number of suspensions was caused by consistent high heat index recorded throughout the country. The Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) recorded the all-time highest heat

index in Metro Manila on April 28, 2024 with a value of 45°C. However, other areas experience worse heat index than Metro Manila with Aparri, Cagayan logging a 48°C heat index followed by Virac, Catanduanes at 47°C (Untalan, 2024). PAGASA classifies heat index of 42-51°C as dangerous where heat cramps and heat exhaustion are likely to happen while heat stroke is also probable with continued exposure.

1.1.2 Urbanization and Environmental Changes

Aside from greenhouse gas emissions, urbanization also plays a pivotal role in driving environmental changes, particularly in the context of land cover change and increased land surface temperatures. As humans aim to expand cities, natural landscapes often fall victim to conversion into built-up areas. This leads to the decrease in vegetation and increase in impervious surfaces. This change has a direct impact on climates, not only globally but also locally, primarily by introducing urban heat island (UHI) effect. UHI makes urban regions experience higher temperatures than their rural counterparts (Stewart & Oke, 2012). Replacing the natural land cover of the Earth with materials such as concrete and asphalt reduces the area's ability to maintain low temperature and cool itself through evapotranspiration and increases the absorption and retention of solar radiation (Mohajerani et al., 2017). Vegetation typically has a higher albedo and can efficiently dissipate heat. In contrast, most urban materials have lower albedos and can store heat which they release slowly over time.

1.1.3 GIS for Environmental Monitoring

Geographic Information Systems (GIS) is an indispensable tool for environmental monitoring. It can be used in the analysis of land surface temperature (LST) and land cover change. By

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integrating spatial data from different sources such as satellite imagery and on-the-ground sensors, GIS can enable researchers to analyze and map the spatial distribution of temperature and land cover (Yang et al., 2002; Onačillová et al., 2022). The ability to map the spatial distribution of both LST and land cover is critical for identifying urban heat islands and assessing impact of land cover changes on local climates. For example, GIS can be used to show the correlation between an increase in LST and expansion of urban areas and a corresponding decrease in vegetation (Ullah et al., 2020). Analyses such as this are crucial for urban planning and strategy development to mitigate the adverse effects of urban heat islands.

1.2 Significance of the Study

By analyzing the spatial and temporal variations of land cover and land surface temperature, this study provides critical insights into the interactions between urban development and local climate patterns. The findings of this study will offer a valuable baseline for assessing how effective the current urban planning policies are in considering their environmental repercussions. The outcomes of this study can inform decision-making processes targeted at adopting resilient urban ecosystems that are designed to withstand the challenges presented by rapid urbanization and climate change.

1.3 Research Objectives

This study focuses on exploring the changes that have happened in the land cover and surface temperature of GCR+. Because of this, the project aims to answer these three questions:

1. How has the surface temperature in the GCR+ changed from 2017 to 2023?
2. How has the land cover in the GCR+ changed from 2017 to 2023?
3. Is there a correlation between the changes in land cover and the changes in surface temperature in GCR+?

Consequently, this project aims to achieve three objectives, each of which addresses the above-mentioned research questions. The specific research objectives are as follows:

1. To analyze the spatial and temporal changes in surface temperature in the GCR+ from 2017 to 2023.
2. To analyze the spatial and temporal changes in land cover in the GCR+ from 2017 to 2023.
3. To investigate the relationship between changes in land cover and changes in surface temperature in GCR+.

2. METHODOLOGY

2.1 Study Area

2.1.1 Greater Capital Region +

In the context of this study, the Greater Capital Region Plus (GCR+) is made of up the National Capital Region (NCR) and its neighboring provinces. These provinces include Cavite, Laguna, Rizal, and Batangas of southern Luzon (Region IV-A); and Bulacan and Pampanga of Central Luzon (Region III). The GCR+ has a total population of 33,596,720 as of 2020 (PSA, 2020). Furthermore, it has a total land area of 13,156.66 square kilometers. Within its borders are two of the country's most famous inland waters: Laguna Lake and Taal Lake, and an active volcano in Taal.



Figure 1 Extent of the GCR+.

These areas collectively play a significant role in Philippine's socio-economic landscape as they represent a hub of governance, commerce, and education. NCR, in particular, is known for its rapid urban growth leading to many environmental challenges including poor water and air quality (Regmi, 2018). In recent years, its neighboring provinces have also seen a sharp increase in urbanization, making it an ideal case study for this project.

2.2 Data

2.2.1 Data Collection

This study used two primary datasets. The first is Environmental Systems Research Institute's (ESRI) Land use/Land Cover dataset that is freely available online and distributed using a Creative Commons by Attribution (CC BY 4.0) license. ESRI describes this dataset as 'high-resolution, open, accurate, comparable, and timely' land cover maps (ESRI, n.d.). These maps were produced using artificial intelligence (AI) land classification models. Sentinel-2 datasets for each year (over 2,000,000 Earth observations from 6 spectral bands) was used to produce the maps. The data is effectively a 9-class raster dataset of the Earth's surface which includes, but is not limited to, the following classes: vegetation types, bare surface, water, cropland and built areas. This dataset has a spatial resolution of 10 meters with a temporal resolution of 1 year from 2017-2023.

The other dataset used in this study was satellite derived LST raster datasets for year 2017-2023. The LST raster contains temperature values in degree Celsius averaged over the entire year. The derivation of LST layers from Landsat image is complex and explained further in the next section. The administrative boundaries used in this study were downloaded from GADM, an online repository of administrative areas of all countries at all levels of sub-divisions.

Table 1 Summary of the datasets in this study

Dataset	Source	Resolution
Land Use/Land Cover	ESRI ArcGIS Living Atlas	10-meter resolution; Annual
Land Surface Temperature	Landsat 8 Imagery; GEE Data Catalog	30-meter resolution; 16-day revisit time
Administrative Boundaries	GADM	N/A

2.2.2 LST Data Preparation

While there are existing land surface temperature (LST) datasets already available online that fit the timeframe of this study, the spatial resolution of such open datasets is low and may not show enough spatial variation for the study area. For this reason, LST data was derived from satellite images, specifically Landsat 8 imagery, to produce higher resolution temperature data. The workflow used to produce LST data is shown in Figure 2. Google Earth Engine (GEE), a remote sensing and cloud-based geospatial analysis platform that enables the user to analyze massive datasets, was used to automate the processing LST data. The Landsat 8 images hosted in GEE contain 5 visible and near-infrared (VNIR) bands, 2 short-wave infrared (SWIR) bands, and one thermal infrared (TIR) band. These bands have already been processed to orthorectified surface reflectance and orthorectified surface temperature before they are served to users.

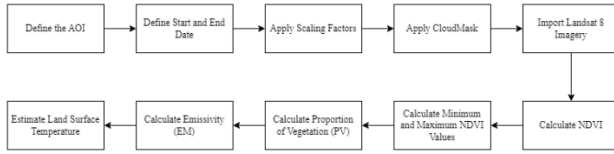


Figure 2 LST estimation workflow (Ridho, 2023)

The first step done in producing LST data from Landsat 8 was to define the area of interest (AOI). A shapefile of the GCR+ administrative boundaries was uploaded to GEE as a new asset. This shapefile was then imported into the script as a Feature Collection. The next step was to define that start and end dates for Landsat 8 imagery retrieval. Because that goal is to produce an LST raster layer for each year, the start and end dates were iteratively changed to the first and last days of the year in subject (i.e., January 1 and December 31, respectively for each year).

GEE typically provides scaling factors and offset coefficients for the satellite images it hosts. Because of this, a 0.0000275 scale and -0.2 offset was applied to the visible and infrared bands while a 0.00341802 and a 149 offset was applied to the thermal infrared band. Cloud masking was also performed to identify and remove cloud-affected pixels from the satellite imagery using the 'QA_PIXEL' band. After these necessary preprocessing techniques, the Landsat 8 imagery is finally imported and used for LST production.

To do the actual LST estimation, the Normalized Difference Vegetation Index (NDVI) was first calculated. NDVI is a popular remote sensing index used in environmental science and vegetation monitoring studies. The NDVI values were calculated using this formula:

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (\text{Equation 1})$$

where:

NDVI = normalized difference vegetation index

NIR = near infrared band of Landsat 8

RED = red band of Landsat 8

After calculating the NDVI, the minimum and maximum NDVI values were then extracted using GEE's reducer function. These minimum and maximum NDVI values were then used to calculate the Proportion of Vegetation (PV). The formula used to calculate PV is:

$$PV = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \right)^2 \quad (\text{Equation 2})$$

where:

PV = proportion of vegetation

NDVI = pixel values of NDVI image

NDVI_{min} = minimum pixel value of NDVI image

NDVI_{max} = maximum pixel value of NDVI image

Then, the Land Surface Emissivity (EM) value is calculated using the following equation:

$$EM = 0.004 * PV + 0.986 \quad (\text{Equation 3})$$

where:

EM = land surface emissivity

PV = proportion of vegetation

NDVI = pixel values of NDVI image

NDVI_{min} = minimum pixel value of NDVI image

NDVI_{max} = maximum pixel value of NDVI image

Finally, the LST values were calculated using the following formula:

$$LST = \left(\frac{BT}{1 + \left(\lambda * \left(\frac{BT}{1.438} \right) * \ln(EM) \right)} \right) - 273.15 \quad (\text{Equation 4})$$

where:

LST = land surface temperature

BT = top-of-atmosphere brightness temperature (in this case the thermal band 10)

λ = wavelength of emitted radiance for Landsat 8 thermal band (11.5um)

EM = land surface emissivity

The result of these steps was a raster layer containing LST values for GCR+ for one year. These steps were written as a GEE script and the start and end dates were changed to acquire LST raster layer for each year.

2.3 Geospatial Processing and Analysis (QGIS)

2.3.1 Preprocessing

After the downloading the ESRI Land Use/Land Cover (LULC) dataset for years 2017-2023 and preparing the Landsat 8 LST raster layers for the same time period, the datasets were then exported to QGIS for further analysis.

Because the ESRI Land Use/Land Cover dataset is tiled and covers a very large area beyond the study area, the ESRI LULC tiles that overlap with the GADM administrative boundaries were clipped and merged using QGIS' geoprocessing tools. These steps were repeated for all years. Both datasets were also reprojected to the same coordinate system (EPSG: 3651).

2.3.2 Raster Calculations

The ESRI LULC data has 9 classes stored in a raster layer. Each land cover class has a unique number value. To identify the land cover change from one year to the next, the raster calculator was used. The cell value of the succeeding year was simply appended to the cell value of the initial year to yield unique values that represented the change from one land cover type to another. This means that if the value 4 is trees and 7 is built-up, if the resulting raster is 47, it means that the cell changed from trees to built-up.

For LST, three unique computations were done. First, all the raster layers were added, and their average was taken using the raster calculator. This calculation represents the LST baseline data for year 2017-2023. Secondly, the resulting LST baseline data done in the first step was subtracted from each of the LST raster layer per year. This effectively gave us the temperature anomaly per year. Finally, temperature change was calculated by subtracting the LST raster layer of the current year from the succeeding year.

2.3.3 Extracting Statistics

To compute the necessary statistics for the LULC, LULC change, LST, baseline LST (average), LST anomalies, and LST change for all the years, the Raster Layer Statistics tool of QGIS was used. In addition, more comprehensive statistics related to the administrative boundaries (provincial level) were extracted using the Zonal Statistics tool of QGIS. This tool appended the statistics to the attribute tables of the provinces which were then exported as CSV files.

2.4 Correlation Analysis and Data Visualization

2.4.1 Correlation Analysis

To understand the trend and relationship between the changes in land cover and temperature, a correlation analysis was performed. For the years 2017-2013, the average LST and area of built-up and tree cover for the whole GCR+ was recorded. The LST was plotted first against the area of built-up in the whole GCR and then against the tree cover area to show how the LST behaves as the built-up area and tree cover increase or decrease. In addition, Python, specifically the SciPy library, was used to calculate the Pearson Correlation Coefficient given by:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (\text{Equation 5})$$

$$\bar{x}A = \pi r^2$$

where:

r = pearson correlation coefficient

x_i = values of the x-variable (area of land cover type)

$\bar{x}$ = mean of the values of the x-variable (area of land cover type)

y_i = values of the y-variable (LST)

$\bar{y}$ = mean of the values of the y-variable (LST)

2.4.2 Data Visualization

Multiple maps were made to show the spatial distribution of the land surface temperature per year in GCR+ as well as the average LST, LST anomalies per, and LST change per year. Similarly, separate maps were produced to show the land cover type in GCR+ per year as well as the land cover changes per year.

Apart from maps, other visualization types such as line plots were produced to visualize trends especially in temperature change. Line plots grouped by land cover types and provinces were also produced for a more comprehensive understanding of the context. Finally, scatterplots with trendlines were created to show the correlation between temperature and land cover.

3. RESULTS AND DISCUSSION

3.1 Land Cover

Mapping the land cover in GCR+ and looking at the spatial distribution of each land cover type, it can be seen that across all the years, tree cover is the most abundant land cover type for almost all provinces with the exception of NCR and Pampanga. The NCR in particular is dominated by urban or built-up areas. Almost 90% of NCR's land cover was classified as built-up. In terms of percentages, Cavite and Pampanga were the 2nd and 3rd in rank for provinces with the highest built-up area percentage (against their respective total land areas).

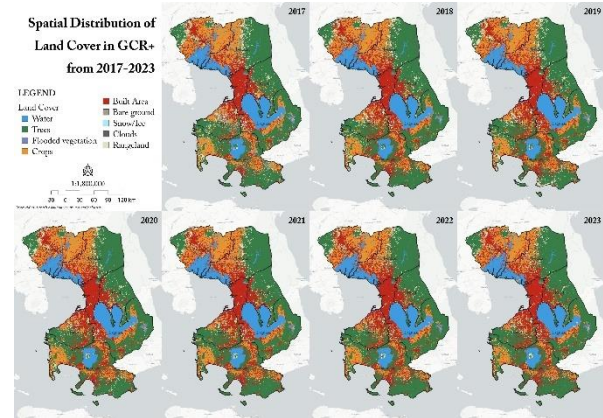


Figure 3 Spatial Distribution of Land Cover Types in GCR+ from 2017-2023.

It can be observed that all provinces have negligible increase in vegetation. However, there is significant increase in built-up areas over the years. Interestingly, data shows that rangeland to tree cover is the most common change especially in the early years of the study period (from 2017-2020). Unsurprisingly, the data also shows that trees to built-up area is one of the most common land cover changes to occur during the study period. The data shows that the change from tree to built-up area is included in the top three land cover changes for almost all the years.

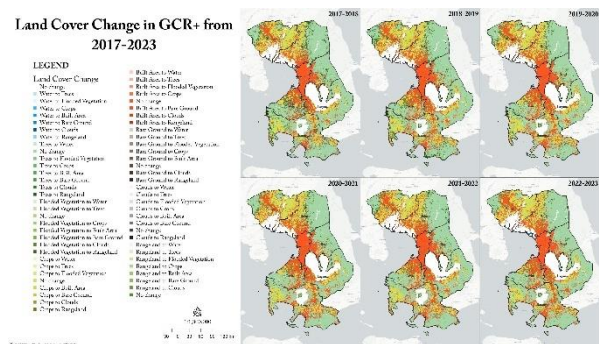


Figure 4 Land Cover Change in GCR+ from 2017-2023.

3.2 Land Surface Temperature

Figure 5 shows the spatial distribution of baseline LST in GCR+ from 2017-2023. The baseline LST is the average LST including all the years in the study period.

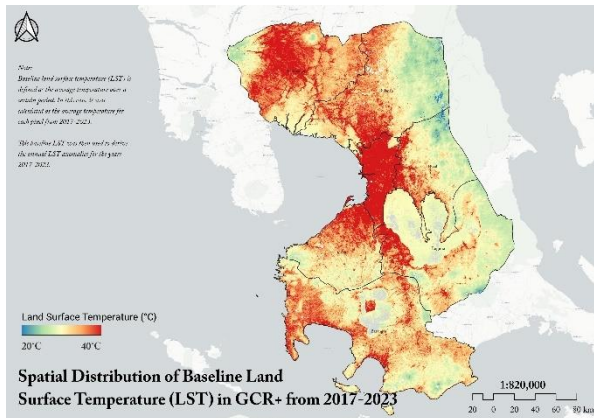


Figure 5 Spatial Distribution of Baseline Land Surface Temperature (LST) in GCR+ from 2017-2023.

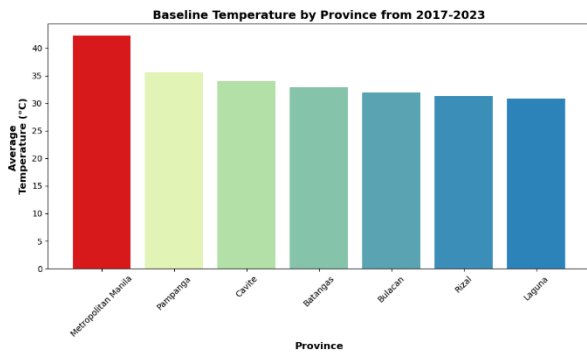


Figure 6 Baseline Temperature by Province from 2017-2023.

It can be seen that the highest temperatures across the entire study period are observed in NCR, recording LST values over 40°C. The second hottest province based on the data is Pampanga, followed by Cavite and Batangas. It is worth noting that the top three hottest provinces (i.e., NCR, Pampanga, and Cavite) also have the most urban areas. It can also be observed in Figure 5 that the eastern part of GCR experiences relatively low temperatures. Looking at the land cover maps, these areas are predominantly vegetation-heavy areas which implies that vegetation cover helps reduce temperatures.

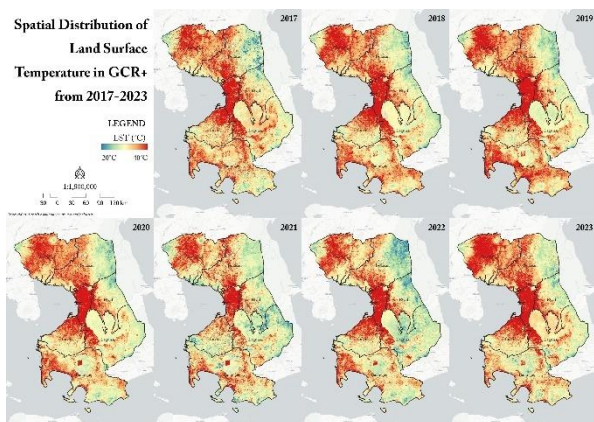


Figure 7 Spatial Distribution of Land Surface Temperature in GCR+ from 2017-2023.

While the general spatial distribution of the temperature values remained the same across the years (NCR, Pampanga, and Cavite being the hottest areas and the eastern portion of GCR experiencing constant low LST values), it was observed that NCR is still the hottest region by a large margin. This may be attributed to the fact that NCR is composed of almost over 90% built-up areas.

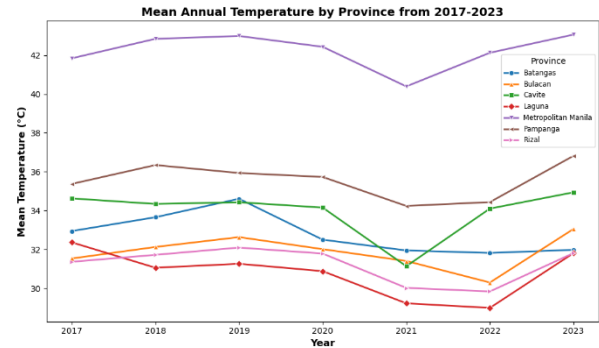


Figure 8 Mean Annual Temperature by Province from 2017-2023.

Looking at the LST anomalies in Figure 9, some interesting events can be seen. In 2017, Cavite, Laguna, and Batangas are the most obvious provinces that experience positive anomaly. Positive anomaly means that they are experiencing higher temperatures than normal. However, come 2018, this is no longer the case. It is now NCR, northern areas of Cavite, and south-eastern areas of Batangas that experience positive LST anomalies. In 2019, almost all provinces experienced positive anomaly values. What this tells us is that during the early years of the study period (pre-2020), generally, the LST anomaly is positive for most provinces.

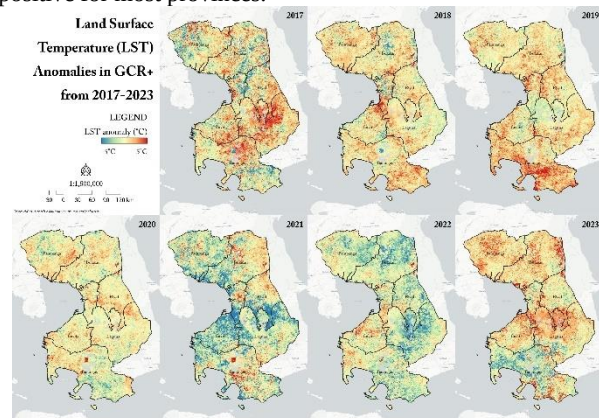


Figure 9 Land Surface Temperature Anomalies in GCR+ from 2017-2023.

The positive anomalies during the early years can be observed better in Figure 10 where steady and gradual increase can be observed for almost all the provinces until the year 2020. In Figure 10, we can also see the most interesting occurrence in this study. That is, the sharp drop in temperature anomaly during 2020 and 2021. This is backed up by the temperature anomaly maps in Figure 9 where almost all the provinces experienced very low LST anomaly values.

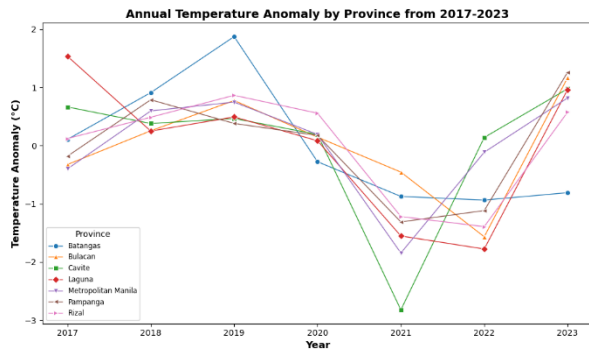


Figure 10 Annual Temperature Anomaly by Province from 2017-2023.

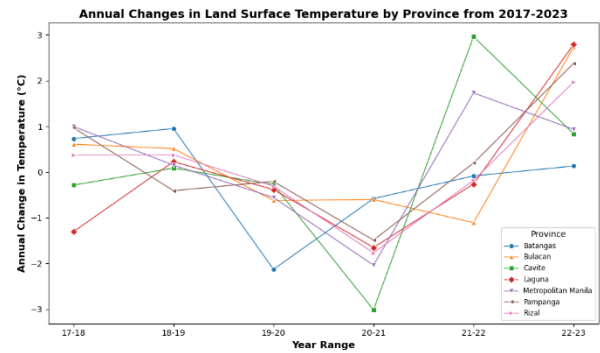


Figure 12 Annual Changes in Land Surface Temperature by Province from 2017-2023.

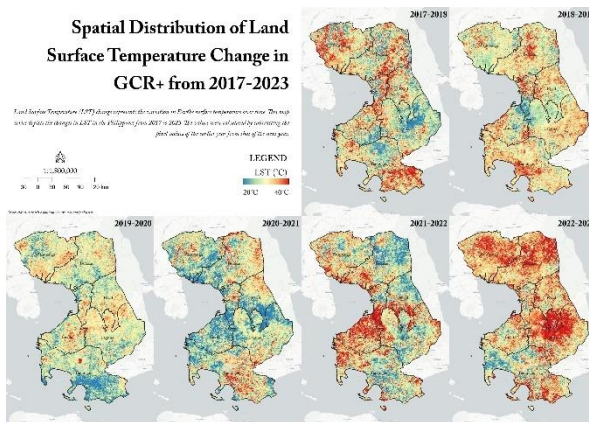


Figure 11 Spatial Distribution of Land Surface Temperature Change in GCR+ from 2017-2023.

This drop in temperature for the years 2020 and 2021 can be observed better in Figure 11 which shows temperature change from year to year. Most notably, we look at the temperature change from 2019-2020 and 2020-2021 and what we observed is that across almost the entire GCR+, there was a sharp decrease in temperatures. Figure 12 summarizes this as a line plot and it is evident that the temperature change in the years 2019-2021 was unusually low. This may have been caused by the COVID-19 pandemic which caused lockdowns over the entire Philippines, of course including GCR+. The abrupt decrease in the carbon emissions and other greenhouse gases may have caused this massive decrease in temperatures. As the country eased up on the travel restrictions starting mid-2022 onwards, the temperature recording has now continued to increase as expected. In fact, the 2022 to 2023 change in temperature is so sharp that is impossible to miss. Even in Figure 11, almost the entire GCR+ is painted red showing that there is a high intensity increase in temperature throughout the region.

Figure 13 shows the temperature change of the difference land cover changes throughout the study period. Land cover changes that are related to bare ground, built-up area, and rangeland consistently show relatively higher (positive) temperature changes. However, it can be seen that, generally, the graph starts relatively high, dips down at the middle, then goes back up towards the end. This trend still follows what we have earlier established where the early years (pre-2020) experienced high temperatures, then the COVID-19 lockdown caused abrupt decrease in temperatures across the region, finally 2023 saw the sharp increase in temperatures again.

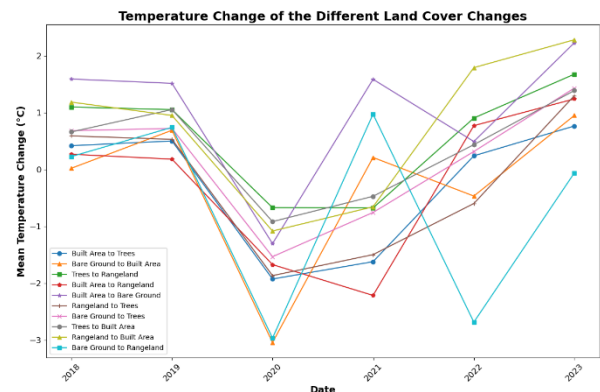


Figure 13 Temperature Change of the Different Land Cover Changes.

By the end of 2023, according to the graph, the changes 'Rangeland to Built Area' and 'Built Area to Bare Ground' showed the highest positive temperature change.

3.3 Correlation Analysis

Figure 14 shows the trend on how LST behaves as built-up area and tree cover increases. This data is for the entire GCR+. As expected, as tree cover increases, the LST value decreases (Figure 14 right). The LST and tree cover area increase resulted in a Pearson Correlation Coefficient value of -0.45 which means that they are negatively correlated (as vegetation increase, temperature decrease). However, interestingly, LST and built-up area showed a similar relationship albeit in a much weaker sense with only a Pearson Correlation Coefficient value of -0.04. This value is almost close to 0 which means that they have no correlation.

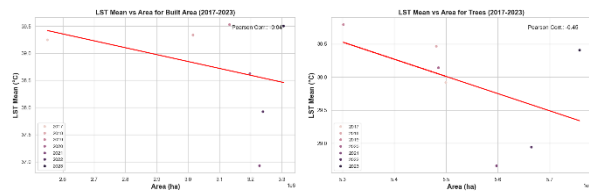


Figure 14 Mean LST vs Built Area (left) & Mean LST vs Trees (right)

Because of the negative correlation of LST and built-up area, we looked at some examples in the provinces and not for the entire GCR+. In Metro Manila, an increase in built-up area and LST showed positive correlation with coefficient of 0.12. However, in Cavite, there is strong negative correlation between the two (coefficient of -0.48). The other provinces also showed varying results. This may imply two things. First is that built-up areas and LST are not correlated or there are other factors not captured within the context of this study.

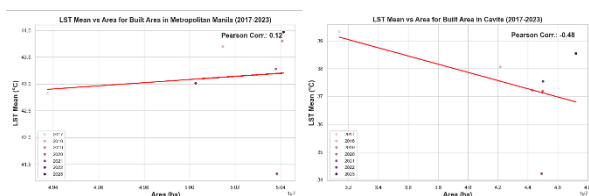


Figure 15 Mean LST vs Built Area in Metro Manila (left) and in Cavite (right)

Similarly, looking at the provincial level for tree cover and LST, similar varying results can also be seen. These are interesting results that should be further studied and understood as there are many probable causes as to why these are happening.

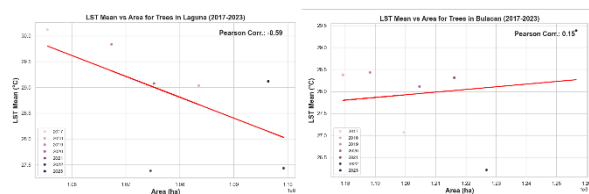


Figure 16 Mean LST vs Tree Cover in Laguna (left) and in Bulacan (right)

4. CONCLUSIONS AND RECOMMENDATIONS

4.1 Conclusions

This study analyzed the spatial and temporal variations of land cover types and land surface temperatures in GCR+. Correlation analyses were also done to understand trends and relationships between the two.

The results of this study showed that tree cover and vegetation, in general, is still the most dominant land cover type across the region. However, tree cover and vegetation are also currently experiencing steady and gradual decline. The provinces of Laguna, Rizal, and Bulacan have the highest tree cover among all the provinces of GCR+. It was also observed that there is a rapid increase in urban areas particularly in Metro Manila, Cavite, and Pampanga. These three regions also have the highest average temperatures throughout the region (although there is a significant difference between Metro Manila and the other two). On the other hand, Laguna, Rizal, and Bulacan experienced the lowest temperatures consistently.

In terms of temperature trends, this study observed that there was a sharp decrease in temperature in 2020 and 2021. This may be caused by the travel restrictions and lockdown which limited human activities related to greenhouse gas emissions. However, 2021 onwards saw a sharp increase in temperatures again. This coincides with the easing up of travel restrictions in the country. It was also observed that built-up areas and bare soil generally exhibit higher temperatures or positive temperature change. Trees are the opposite as they exhibit relatively low temperatures. Finally, this study concludes that changes in land cover do not completely explain variations in temperature; however, general trends can be observed.

Modelling environmental systems is difficult because it is complex and there are a lot of factors to consider. Similarly, studying temperature is not as straightforward as relating it to land cover. However, there are insights to be derived from studying these two together that can help us make better decisions. We hope that future studies on temperature and its relationship with other environmental variables such as atmospheric gases are explored to complement this research. By doing so, we can understand a bit better the effects of whatever action we make as humans.

4.2 Recommendations

The author recommends to future researchers to expand the scope of the study and/or include other urbanized regions such as Davao or Iloilo. It is also very important and possible to incorporate more explanatory variables that influence land surface temperature. If available, it is recommended to use higher resolution datasets used in this study to capture more accurately what is true on the ground. If possible, it is ideal to extend the time period of the analysis to observe trends better. In addition, considering seasonal variations in this type of study will be useful to provide additional context. Future research can also incorporate validation of the results. Finally, the author recommends exploring more advanced statistical analyses.

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